Homework — 2.1.2 Thinking Ahead

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 2.1.2 lesson.

Part A — This week's topic (~14 marks)

1. A library system has a routine that issues a book to a member.

(a) Identify **one** input and **one** output of this routine. **[2]** *(AO2)*

(b) "Check the member has no unpaid fines" is part of this routine. State whether it is an input, an output or a process, and justify your answer. **[1]** *(AO2)*

2. Before a routine can mark a member's book as "returned", the system must first confirm the member actually has that book on loan.

State the **precondition** for the "return book" routine and explain what could go wrong if the routine ran without checking it. **[2]** *(AO2)*

3. A train-times website looks up the same popular "London → Manchester" timetable for thousands of users every hour. The developers decide to **cache** the computed timetable.

(a) Explain **one** benefit of caching in this scenario. **[2]** *(AO2)*

(b) Explain **one** trade-off, referring to what happens if a train is delayed. **[2]** *(AO2)*

4. A games studio is building three different games that each need a feature to load and save the player's progress to a file. Explain **two** benefits of writing this as a single **reusable component** rather than coding it separately in each game. **[2]** *(AO2)*

5. A developer is planning a routine that processes an online food order. Using *thinking ahead*, state for this routine: **one** input, a **precondition** that must hold before payment is taken, and **one** thing it would be useful to **cache**. **[3]** *(AO2)*

Part B — Retrieval: keep it fresh (~6 marks)

6. *(2.1.1)* State **one** detail a developer would **keep** and **one** they would **discard** when abstracting a "student" for a school timetabling system. **[2]** *(AO2)*

7. *(1.2.1)* State what an **interrupt** is and one reason an operating system uses interrupts. **[2]** *(AO1)*

8. *(1.1.1)* State **two** factors that affect the **performance** of a processor. **[2]** *(AO1)*